Extended Reality
Final Project Guideline

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Overview

The final project is a team-based XR development project where you will design and implement a novel AR, VR, or MR system. Projects may focus on **technical research, system development, or application design**, but they must demonstrate meaningful use of XR technologies.

Each team will:

- Select a project topic
- Submit a short proposal
- Implement a working prototype
- Present the project
- Submit a final report and video demonstration

Projects are not expected to reproduce state-of-the-art research from scratch. Teams may build on existing open-source implementations and focus on **integration, experimentation, or application development**. Projects may be implemented using **Unity, Unreal Engine, WebXR, or other XR frameworks**, depending on the project goals.

Suggested topics

The following are the suggested topics. You can choose one of them, but we impose a rule that no more than three teams can work on the same topic. If more than three teams express their interests in the same topic, we will resolve the conflict on a first-come-first-serve basis using the date and time when the project registration is done.

If you are not interested in the suggested topics, you are welcome to propose a new one. You must discuss the topic with the lecturers and get approval to proceed with your custom project.

Projects

1) *Photorealistic reconstruction 3D environment from videos using 3D Gaussian Splatting*. In this project, you will explore 3D Gaussian Splatting for 3D scene reconstruction for VR. You will apply this method to scan at least three scenes including an indoor scene and an outdoor scene, and the TCD campus scene. You will render the

Gaussians in real time and display the rendering on a VR headset. User should be able to explore the scene while viewing it through the VR headset. Extend the reconstruction to support dynamic scenes (e.g., moving cars, walking pedestrians), if possible.

2) *Talking heads*. In this project, you will implement a method that automatically creates an animated 3D facial avatar from an input image or video. You will integrate at least two avatars into a VR environment, and make the avatars talk to each other.
Reference: Meta's Codec Avatars.

3) *Hand-object interactions*. In this project, you will implement a method for reconstructing hands and the objects that the hands interact with. The reconstruction could include the hand and object meshes, and the hand and object poses. Apply this method in AR or VR to demonstrate how hands could be used to interact with virtual objects. Note: this could be an extension from Assignment 2 and it is expected that the team work would lead to significant changes and improvement.

4) *Full-body reconstruction*. In this project, you will implement a method to estimate full-body geometry and/or poses from an input (ego-centric or regular) video. Apply the resulting reconstruction for avatar creation, for controlling virtual characters, or object interactions in AR/VR.

5) *Segment and track everything*. In this project, you will implement a system that scans and reconstructs the scene in real-time, following dense reconstruction as KinectFusion style, or sparse reconstruction following SLAM. The system then detects, segments, and tracks multiple 3D objects by leveraging a 2D image/video-based segmentation method and lifting the segments to 3D. The system should work with generic object in different categories. The system should allow user to provide a text, image, click, or gesture prompt to query about an object in the scene. The system should provide an answer to the query. Example: in a living room, when user queries "the sofa", the corresponding object should be segmented.

6) *Make it home*. In this project, you will implement a system that performs in-situ 3D content generation for a real empty space using user prompts like text and voice. The system will generate 3D furniture, objects, and decorations that furnish the empty space virtually. Integrate image/video/3D generation models to conceptualize the scene. The system should seamlessly integrate the virtual objects with matching lighting and shadow conditions.

7) *VR navigation*. In this project, you will implement an application that allows users to navigate real-world locations in VR, similar to Google Street Views or Apple Look Around. The application should allow smooth navigation with 360-degree views. Integrate an LLM or VLM to allow users to query the location and relevant scene features (site names, history, etc.)

8) *XR games*. In this project, you will build an AR/VR/MR game with passthrough, demonstrating the interaction between the virtual and the real world. Deploy the game to a headset.

9) *Social VR*. In this project, you will create a social VR virtual environment on Horizon Worlds, VRChat, etc. You will demonstrate the diversity and complexity of the environment you create through at least three scenes, each with its own theme, context, and story.

What makes a strong project

Projects will be evaluated based on:

- Originality – is there a creative or interesting idea?
- Technical implementation – does the system actually work?
- XR relevance – does the project meaningfully use AR/VR/MR technology?
- Presentation quality – clarity of report and demo

Project registration

Note that you will be assigned a team if you do not immediately let us know if you have formed your own team. Please let us know if you have not been allocated to a team. If you do not express interest in any suggested topics or your idea, your team will be assigned to a (random) topic in the recommended list.

If you do not register for a project, the lecturers will assign you to a team, and you will be given a (random) topic from the suggested list.

Please email Gareth.Young@TCD.ie and Binhson.Hua@TCD.ie for project registration. Include the following details in the email.
- Title: [XR project registration] <Your topic>
- Team members
- Additional explanations if your team propose a new topic. Justify if the number of members in your team is more than three.

Project proposal

The proposal should demonstrate that the team has a clear technical plan and realistic scope. The proposal is limited to 1 page or 500-700 words in the ACM format. Example proposal titles:

- "AR Plant Identification System for Campus Spaces"

- “Lighting-Aware AR Furniture Placement”
- “Gaussian Splat Reconstruction of Trinity Campus“

Suggested structure:

- Abstract (~150 words)
- Introduction and motivation
- Related technologies or prior work
- Proposed system
- Implementation plan
- Team responsibilities

The project proposal and scope will be marked, so you must work on the proposal with your team to ensure everyone agrees on the topic, the technical contribution, and the workload for each project member. You can repurpose the proposal to write the abstract and introduction of your final report.

Package the PDF and submit your proposal to Blackboard. Remember to include the names of your team and team members in the PDF.

Project Report

Your project report should follow the structure of a technical report or scientific paper, e.g., as follows:

- Abstract
- Introduction
- Related work / Background
- Method / Implementation
- Experimental Results
- Discussion and Conclusion
- References

The report is limited to a maximum of 4 pages.

For the project proposal and report, it is recommended that you use Overleaf and LaTeX and follow the ACM SIGGRAPH template to write your report.

- A template on Overleaf is provided here: https://www.overleaf.com/read/vtbyjvngrzgz#e28726
- Trinity College Dublin provides professional Overleaf subscriptions for staff and students. https://www.overleaf.com/edu/tcd

Deliverables

Package the following: (0) your proposal, (1) your code, (2) the project final report in PDF, and (3) an illustrative video in mp4 into project-final.zip and submit it to Blackboard.

The video should illustrate the method and main results of your work. The video should show:

- the problem being addressed
- the system in action
- key technical features
- example results

Screen capture with narration is acceptable. By default, we will upload videos to a YouTube playlist for reference for future classes. You can opt out if you do not want your work published on YouTube by emailing the lecturers.

Timeline

- Project registration deadline: Friday, Mar 13, 2026 at 23:59 (midnight).
- Project proposal deadline: Monday, Mar 23, 2026, at 23:59 (midnight).
- Project presentation: **Wednesday, Apr 08, 2026 at 11:00 O'Reilly foyer**.
- Project final submission: **Friday, Apr 10, 2026 at 23:59 (midnight)**.
- Late submissions will be penalized by 10% of the total marks, so your maximum marks for the project will be 50% due to late submissions.
- No further submissions are accepted after **Apr 10, 2026**.

Evaluation criteria

The project is **60%** of the total module marks with the following breakdown:

- Originality, creativity: 10%
- Technical quality and complexity: 30%
- XR relevance, robustness: 10%
- Proposal and final report: 10%.

Past projects

Here is the YouTube playlist of the previous class projects for your reference.

https://www.youtube.com/playlist?list=PL6FW1m-qiD7CY7k967XUlkBm-wSClp8td

https://www.youtube.com/playlist?list=PL6FW1m-qiD7CvuY0drSdREAkD-ucttWV1